Determinants of Matrices

Lecture Notes

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Introduction

The determinant is one of the most important scalar quantities associated with a square matrix. It encodes valuable information about the matrix and the linear transformation it represents. The determinant appears in numerous applications across mathematics, physics, and engineering, from computing areas and volumes to solving systems of linear equations, finding eigenvalues, and analyzing the invertibility of matrices.

In this chapter, we will:

- Define the determinant rigorously using cofactors and minors
- Develop efficient methods for computing determinants
- Explore the key properties and applications of determinants
- Work through detailed examples for various matrix sizes
- Understand why determinants are fundamental to linear algebra

Definition and Basic Concepts

Minors and Cofactors

Definition: Minor

Let A be an $n \times n$ matrix. The (i, j)-**minor** of A, denoted M_{ij}, is the determinant of the $(n-1) \times (n-1)$ submatrix obtained by deleting the i-th row and j-th column of A.

Definition: Cofactor

The (i, j)-**cofactor** of A, denoted C_{ij}, is defined as

$$C_{ij} = (-1)^{i+j} M_{ij}$$

The factor $(-1)^{i+j}$ is called the **sign factor** or **checkerboard pattern**.

Remark: Checkerboard Pattern

The sign factor $(-1)^{i+j}$ creates a checkerboard pattern of signs:

$$\begin{pmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

This pattern ensures that cofactors are correctly signed for the cofactor expansion formula.

The Determinant Definition

Definition: Determinant via Cofactor Expansion

Let A be an $n \times n$ matrix. The **determinant of** A, denoted $\det(A)$ or $|A|$, is defined as

$$\det(A) = \sum_{j=1}^{n} a_{1j} C_{1j} = \sum_{j=1}^{n} a_{1j} (-1)^{1+j} M_{1j}$$

where a_{1j} is the $(1, j)$ entry of A, and the sum is over the first row. This is called **cofactor expansion along the first row**.

More generally, we can expand along any row i or any column j:

$$\det(A) = \sum_{j=1}^{n} a_{ij} C_{ij} \quad \text{(expansion along row } i\text{)}$$

$$\det(A) = \sum_{i=1}^{n} a_{ij} C_{ij} \quad \text{(expansion along column } j\text{)}$$

Remark: Independence of Expansion

A fundamental property of the determinant is that the value is independent of which row or column we choose for expansion. All choices yield the same result. This allows us to choose the row or column with the most zeros to simplify computation.

Computing Determinants: 2 by 2 Matrices

For 2×2 matrices, the determinant formula is simple and can be memorized.

Proposition: 2 by 2 Determinant Formula

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\det(A) = ad - bc$$

Proof

Using cofactor expansion along the first row:

$$\begin{aligned} \det(A) &= a \cdot C_{11} + b \cdot C_{12} \\ &= a \cdot (-1)^{1+1} M_{11} + b \cdot (-1)^{1+2} M_{12} \\ &= a \cdot d + b \cdot (-c) \\ &= ad - bc \end{aligned}$$

Example: Determinant of a 2 by 2 Matrix

Find the determinant of $A = \begin{pmatrix} 3 & 2 \\ 5 & 4 \end{pmatrix}$.

Using the formula:

$$\det(A) = (3)(4) - (2)(5) = 12 - 10 = 2$$

Example: 2 by 2 Matrix with Zero Determinant

Find the determinant of $B = \begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}$.

Notice that the second row is $\frac{3}{2}$ times the first row (the rows are linearly dependent).

$$\det(B) = (2)(6) - (4)(3) = 12 - 12 = 0$$

The determinant is zero, confirming that the matrix is singular (not invertible).

Example: 2 by 2 Rotation Matrix

Find the determinant of the rotation matrix $R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$$\det(R) = (\cos\theta)(\cos\theta) - (-\sin\theta)(\sin\theta) = \cos^2\theta + \sin^2\theta = 1$$

This shows that rotation matrices have determinant 1, which is expected since they represent rotations that preserve area and orientation.

Computing Determinants: 3 by 3 Matrices

For 3×3 matrices, we can use either cofactor expansion or the Rule of Sarrus.

Rule of Sarrus

Definition: Rule of Sarrus

For a 3×3 matrix $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, the determinant can be computed as

$$\det(A) = aei + bfg + cdh - ceg - afh - bdi$$

Alternatively, this can be visualized by:

1. Writing the matrix and repeating the first two columns on the right
2. Computing the product of the three diagonals going down-right (add these products)
3. Computing the product of the three diagonals going down-left (subtract these products)

Example: 3 by 3 Determinant Using Rule of Sarrus

Find the determinant of $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{pmatrix}$.

Using the Rule of Sarrus:

$$\begin{matrix} 1 & 2 & 3 & 1 & 2 \\ 0 & 4 & 5 & 0 & 4 \\ 1 & 0 & 6 & 1 & 0 \end{matrix}$$

Down-right diagonals: $(1)(4)(6) + (2)(5)(1) + (3)(0)(0) = 24 + 10 + 0 = 34$

Down-left diagonals: $(3)(4)(1) + (1)(5)(0) + (2)(0)(6) = 12 + 0 + 0 = 12$

Therefore:

$$\det(A) = 34 - 12 = 22$$

Cofactor Expansion for 3 by 3 Matrices

Example: 3 by 3 Determinant Using Cofactor Expansion

Find the determinant of $B = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 5 \\ 0 & 1 & 2 \end{pmatrix}$ using cofactor expansion.

Since the first column has two zeros, expanding along the first column is most efficient:

$$\begin{aligned} \det(B) &= 2 \cdot C_{11} + 0 \cdot C_{21} + 0 \cdot C_{31} \\ &= 2 \cdot (-1)^{1+1} M_{11} \\ &= 2 \cdot \begin{vmatrix} 4 & 5 \\ 1 & 2 \end{vmatrix} \\ &= 2 \cdot (4 \cdot 2 - 5 \cdot 1) \\ &= 2 \cdot (8 - 5) \\ &= 2 \cdot 3 \\ &= 6 \end{aligned}$$

Example: 3 by 3 Matrix with All Nonzero Entries

Find the determinant of $C = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$.

Notice that the rows are linearly dependent (row 3 = row 2 + (row 2 − row 1), approximately).

Using cofactor expansion along the first row:

$$\begin{aligned} \det(C) &= 1 \cdot \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2 \cdot \begin{vmatrix} 4 & 6 \\ 7 & 9 \end{vmatrix} + 3 \cdot \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} \\ &= 1(45 - 48) - 2(36 - 42) + 3(32 - 35) \\ &= 1(-3) - 2(-6) + 3(-3) \\ &= -3 + 12 - 9 \\ &= 0 \end{aligned}$$

As expected, the determinant is zero because the rows are linearly dependent.

Computing Determinants: n by n Matrices

General Cofactor Expansion Method

For an $n \times n$ matrix, cofactor expansion works for any n, but it becomes computationally expensive for large n due to the recursive nature of the calculation.

Example: 4 by 4 Determinant Using Cofactor Expansion

Find the determinant of

$$A = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 3 & 1 & 1 & 2 \\ 0 & 4 & 1 & 3 \\ 2 & 0 & 1 & 0 \end{pmatrix}$$

Expanding along the first row (which has two zeros):

$$\begin{aligned} \det(A) &= 1 \cdot C_{11} + 0 \cdot C_{12} + 2 \cdot C_{13} + 0 \cdot C_{14} \\ &= 1 \cdot (-1)^{1+1} M_{11} + 2 \cdot (-1)^{1+3} M_{13} \end{aligned}$$

For M_{11}, delete row 1 and column 1:

$$M_{11} = \begin{vmatrix} 1 & 1 & 2 \\ 4 & 1 & 3 \\ 0 & 1 & 0 \end{vmatrix}$$

Expanding along the third row:

$$M_{11} = 0 - 1 \cdot (-1)^{3+2} \begin{vmatrix} 1 & 2 \\ 4 & 3 \end{vmatrix} + 0 = -1(-1)(3-8) = -1(-1)(-5) = -5$$

For M_{13}, delete row 1 and column 3:

$$M_{13} = \begin{vmatrix} 3 & 1 & 2 \\ 0 & 4 & 3 \\ 2 & 0 & 0 \end{vmatrix}$$

Expanding along the third row:

$$M_{13} = 2 \cdot (-1)^{3+1} \begin{vmatrix} 1 & 2 \\ 4 & 3 \end{vmatrix} = 2(3-8) = 2(-5) = -10$$

Therefore:

$$\det(A) = 1 \cdot 1 \cdot (-5) + 2 \cdot 1 \cdot (-10) = -5 - 20 = -25$$

Row Reduction Method

An efficient method for computing determinants of large matrices is to use row reduction, keeping track of how row operations affect the determinant.

Proposition: Effect of Row Operations on Determinant

When applying elementary row operations to a matrix A:

(i) If row i is multiplied by a scalar c, then $\det(A_{\text{new}}) = c \cdot \det(A)$

(ii) If two rows are swapped, then $\det(A_{\text{new}}) = -\det(A)$

(iii) If a multiple of one row is added to another row, then $\det(A_{\text{new}}) = \det(A)$ (determinant is unchanged)

Example: Computing Determinant via Row Reduction

Find the determinant of

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 4 & 2 & 6 \\ 1 & 0 & 2 \end{pmatrix}$$

Step 1: Swap rows 1 and 3 (determinant changes sign)

$$\begin{pmatrix} 1 & 0 & 2 \\ 4 & 2 & 6 \\ 2 & 1 & 3 \end{pmatrix}, \quad \det = -\det(A)$$

Step 2: $R_2 \to R_2 - 4R_1$ and $R_3 \to R_3 - 2R_1$ (determinant unchanged)

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 2 & -2 \\ 0 & 1 & -1 \end{pmatrix}, \quad \det = -\det(A)$$

Step 3: Swap rows 2 and 3 (determinant changes sign)

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 2 & -2 \end{pmatrix}, \quad \det = \det(A)$$

Step 4: $R_3 \to R_3 - 2R_2$ (determinant unchanged)

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \det = \det(A)$$

The upper triangular matrix has a zero on the diagonal, so its determinant is 0. Therefore, $\det(A) = 0$.

Determinant of Upper Triangular Matrices

Proposition: Determinant of Triangular Matrices

For an upper triangular matrix $A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}$,

$$\det(A) = a_{11} \cdot a_{22} \cdot a_{33} \cdots a_{nn}$$

That is, the determinant is the product of the diagonal entries.

Example: Determinant of an Upper Triangular Matrix

Find the determinant of

$$U = \begin{pmatrix} 3 & 2 & 4 & 1 \\ 0 & -1 & 5 & 2 \\ 0 & 0 & 2 & 3 \\ 0 & 0 & 0 & 4 \end{pmatrix}$$

Since U is upper triangular:

$$\det(U) = 3 \cdot (-1) \cdot 2 \cdot 4 = -24$$

Properties of Determinants

Theorem: Fundamental Properties of Determinants

Let A and B be $n \times n$ matrices. Then:

(i) $\det(AB) = \det(A) \cdot \det(B)$ (multiplicative property)

(ii) $\det(A^T) = \det(A)$ (transposition preserves determinant)

(iii) If A has a row or column of zeros, then $\det(A) = 0$

(iv) If A has two identical rows or columns, then $\det(A) = 0$

(v) If A has two proportional rows or columns, then $\det(A) = 0$

(vi) $\det(cA) = c^n \det(A)$ for any scalar c (scaling all entries)

(vii) If A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$

Example: Product Rule for Determinants

Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$.
We can verify the product rule:

$$\begin{aligned} \det(A) &= 1(4) - 2(3) = 4 - 6 = -2 \\ \det(B) &= 2(3) - 0(1) = 6 \\ \det(A) \cdot \det(B) &= (-2)(6) = -12 \end{aligned}$$

Computing AB:

$$AB = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 10 & 12 \end{pmatrix}$$

$$\det(AB) = 4(12) - 6(10) = 48 - 60 = -12 \quad \checkmark$$

Example: Invertibility Test

Determine which matrices are invertible:
(a) $A = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}$: $\det(A) = 2(2) - 4(1) = 0$. Not invertible.
(b) $B = \begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$: $\det(B) = 2(3) - 4(1) = 6 - 4 = 2 \neq 0$. Invertible.
If B is invertible, then $\det(B^{-1}) = \frac{1}{\det(B)} = \frac{1}{2}$.

Applications and Importance of Determinants

Invertibility of Matrices

Theorem: Invertibility Criterion

An $n \times n$ matrix A is invertible if and only if $\det(A) \neq 0$.

Remark: Relationship to Rank

For an $n \times n$ matrix A:

- If $\det(A) \neq 0$, then $\text{rank}(A) = n$ (full rank)
- If $\det(A) = 0$, then $\text{rank}(A) < n$ (deficient rank)
- The rows and columns are linearly independent if and only if $\det(A) \neq 0$

Geometric Interpretation

Definition: Geometric Meaning of Determinant

For an $n \times n$ matrix A whose column vectors are $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \in \mathbb{R}^n$:

- In $\mathbb{R}^2$: $|\det(A)|$ is the area of the parallelogram formed by the two column vectors
- In $\mathbb{R}^3$: $|\det(A)|$ is the volume of the parallelepiped formed by the three column vectors
- In $\mathbb{R}^n$: $|\det(A)|$ is the n-dimensional volume of the parallelepiped formed by the column vectors
- The sign of $\det(A)$ indicates orientation: positive means the transformation preserves orientation, negative means it reverses orientation

Example: Area of a Parallelogram

Find the area of the parallelogram formed by vectors $\mathbf{u} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$.

The area is:

$$\text{Area} = \left| \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} \right| = |3(4) - 1(1)| = |12 - 1| = 11$$

Example: Volume of a Parallelepiped

Find the volume of the parallelepiped formed by vectors $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$, and $\mathbf{w} = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$.

The volume is:

$$\text{Volume} = \left| \begin{vmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{vmatrix} \right| = |1 \cdot 2 \cdot 3| = 6$$

Cramer's Rule

Theorem: Cramer's Rule

Let A be an $n \times n$ invertible matrix and let $\mathbf{b}$ be a column vector. The unique solution to the system $A\mathbf{x} = \mathbf{b}$ is

$$x_i = \frac{\det(A_i)}{\det(A)}$$

where A_i is the matrix obtained by replacing the i-th column of A with $\mathbf{b}$.

Example: Solving a System Using Cramer's Rule

Solve the system using Cramer's rule:

$$2x + y = 5$$
$$3x + 2y = 8$$

In matrix form: $A\mathbf{x} = \mathbf{b}$ where $A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ 8 \end{pmatrix}$.

First, compute $\det(A) = 2(2) - 1(3) = 4 - 3 = 1 \neq 0$ (so unique solution exists).

For x: $A_1 = \begin{pmatrix} 5 & 1 \\ 8 & 2 \end{pmatrix}$, so $\det(A_1) = 5(2) - 1(8) = 10 - 8 = 2$

$$x = \frac{\det(A_1)}{\det(A)} = \frac{2}{1} = 2$$

For y: $A_2 = \begin{pmatrix} 2 & 5 \\ 3 & 8 \end{pmatrix}$, so $\det(A_2) = 2(8) - 5(3) = 16 - 15 = 1$

$$y = \frac{\det(A_2)}{\det(A)} = \frac{1}{1} = 1$$

Therefore, $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Verification: $2(2) + 1 = 5$ ✓ and $3(2) + 2(1) = 8$ ✓

Eigenvalues and the Characteristic Polynomial

Definition: Characteristic Polynomial

The **characteristic polynomial** of an $n \times n$ matrix A is defined as

$$p(\lambda) = \det(A - \lambda I)$$

where I is the $n \times n$ identity matrix and λ is a variable.

The eigenvalues of A are the roots of the characteristic polynomial: $\det(A - \lambda I) = 0$.

Example: Finding Eigenvalues Using the Characteristic Polynomial

Find the eigenvalues of $A = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$.
The characteristic polynomial is:

$$p(\lambda) = \det \begin{pmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{pmatrix} = (3-\lambda)(2-\lambda) - 0 = (3-\lambda)(2-\lambda)$$

Setting $p(\lambda) = 0$:

$$(3-\lambda)(2-\lambda) = 0$$

The eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 2$.

Summary

Determinants are a fundamental concept in linear algebra with far-reaching applications:

- The **definition via cofactors and minors** provides a recursive way to compute determinants for any $n \times n$ matrix
- For 2×2 matrices, the formula $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$ is simple and memorizable
- For 3×3 matrices, we can use either the Rule of Sarrus or cofactor expansion
- For larger matrices, row reduction tracking is often the most efficient computational method
- The **key properties** of determinants—particularly the product rule $\det(AB) = \det(A)\det(B)$ and the sign reversals under row swaps—are essential for both theory and computation
- A matrix is invertible if and only if $\det(A) \neq 0$
- Geometrically, $|\det(A)|$ represents the volume scaling factor of the linear transformation represented by A
- Determinants are essential for understanding eigenvalues, solving systems of equations (Cramer's rule), and numerous other applications in science and engineering

The determinant truly is one of the most important scalar quantities in linear algebra, deserving careful study and practice in computation.